

Lab 2: Implementation of Multiplexer and Encoder

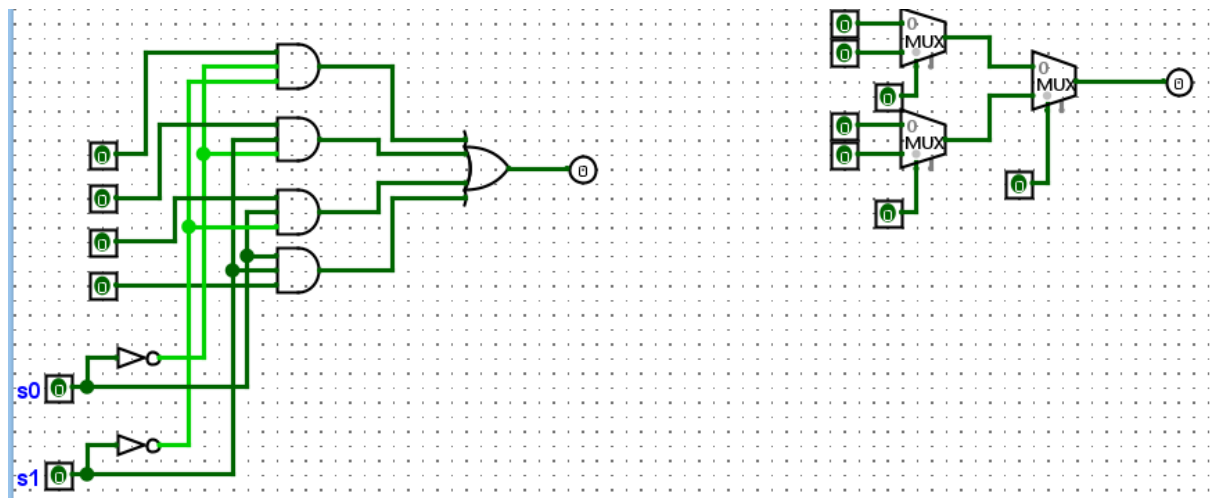
Aim

To design and simulate 4:1 Multiplexer, 8:1 Multiplexer, and 8:3 Encoder using Logisim Evolution and verify their outputs.

Procedure

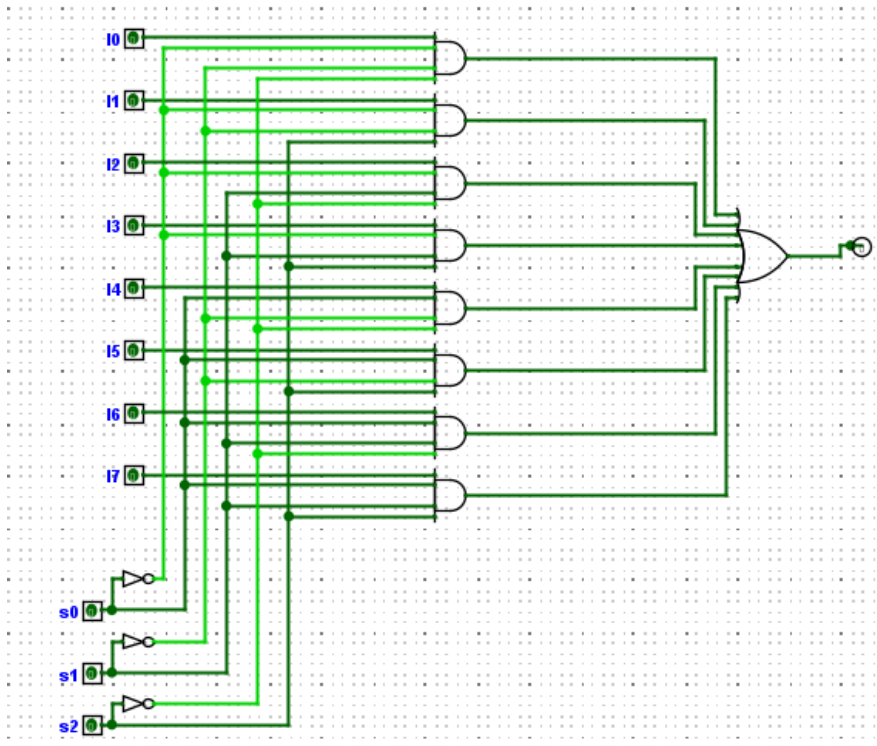
Part A: 4:1 Multiplexer

1. Open Logisim Evolution.
2. Place four input pins (I0, I1, I2, I3) and two select lines (S0, S1).
3. Design the circuit using logic gates or use a built-in MUX component.
4. Connect select lines to control input selection.
5. Connect output to an LED.
6. Apply different input combinations and verify output based on select lines.



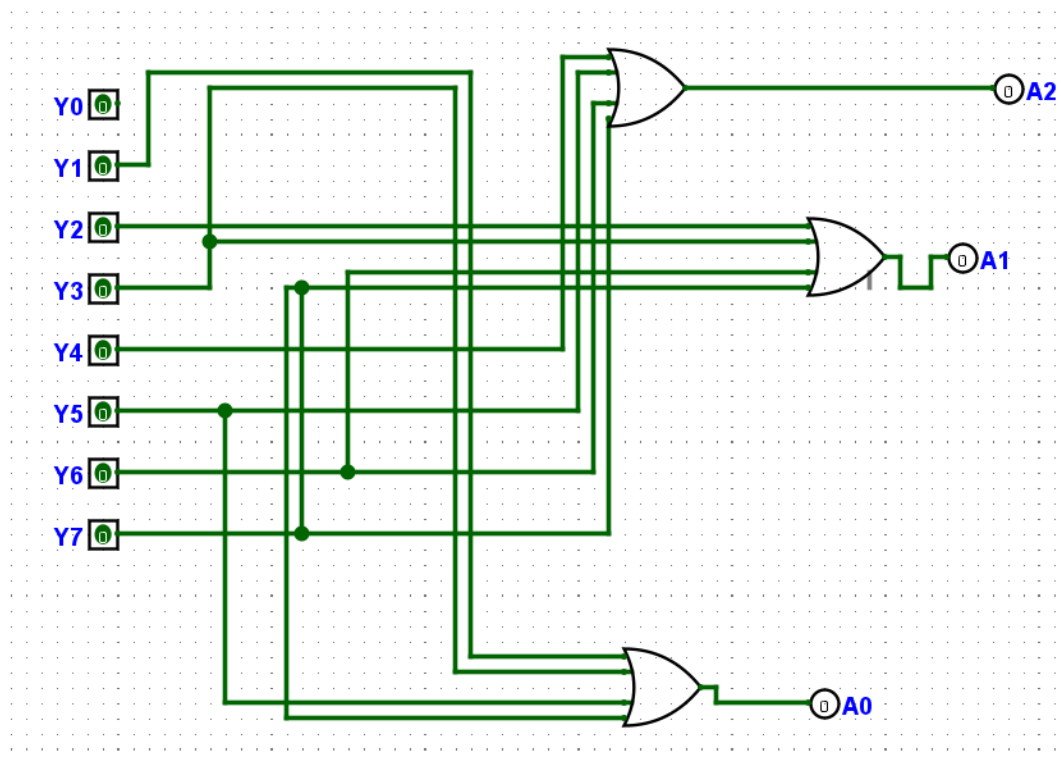
Part B: 8:1 Multiplexer

1. Place eight input pins (I0 to I7) and three select lines (S0, S1, S2).
2. Construct the circuit using logic gates or cascading smaller multiplexers.
3. Connect select lines properly to choose one of the eight inputs.
4. Connect output to LED.
5. Test the circuit with various input combinations and observe output.



Part C: 8:3 Encoder

1. Place eight input pins (D0 to D7).
2. Design the encoder circuit to convert active input into a 3-bit binary output.
3. Use logic gates to generate output bits (Y0, Y1, Y2).
4. Connect outputs to LEDs.
5. Activate one input at a time and observe corresponding binary output.



Result

The 4:1 Multiplexer, 8:1 Multiplexer, and 8:3 Encoder were successfully designed and simulated using Logisim Evolution.

- The 4:1 and 8:1 multiplexers correctly selected inputs based on select lines.
- The 8:3 encoder produced correct binary outputs for corresponding active inputs.

All outputs were verified and matched the expected theoretical results.